

Week 3

# Limits of functions pt 2

## Examples

$$\textcircled{1} \lim_{x \rightarrow 3} (x^2 - 2x) = 9 - 6 = 3$$

$$\textcircled{2} \lim_{x \rightarrow a} \frac{x - a}{\frac{1}{x} - \frac{1}{a}}$$

$$= \lim_{x \rightarrow a} \frac{x - a}{\frac{a - x}{ax}}$$

$$= \lim_{x \rightarrow a} (x - a) \cdot \frac{ax}{a - x} = \lim_{x \rightarrow a} \cancel{(x - a)} \cdot \frac{ax}{-\cancel{(x - a)}} = \lim_{x \rightarrow a} ax = -a^2$$

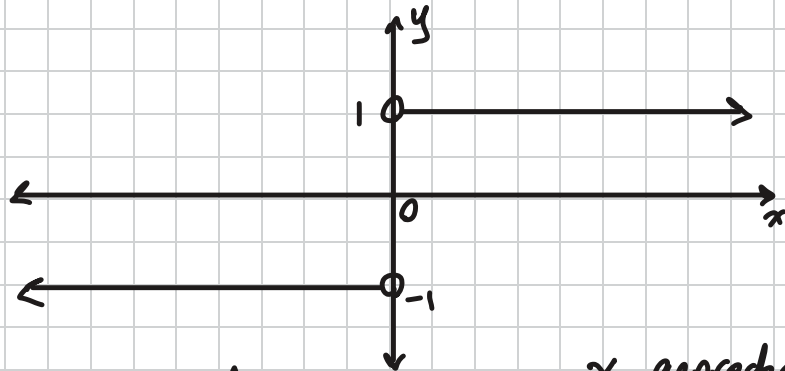
$$\begin{aligned}
 & \textcircled{3} \quad \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}} \\
 &= \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}} \cdot \frac{\sqrt{4+t} + \sqrt{4-t}}{\sqrt{4+t} + \sqrt{4-t}} \\
 &= \lim_{t \rightarrow 0} \frac{t(\sqrt{4+t} + \sqrt{4-t})}{4+t - (4-t)} \\
 &= \lim_{t \rightarrow 0} \frac{\cancel{t}(\sqrt{4+t} + \sqrt{4-t})}{2\cancel{t}} \\
 &= \lim_{t \rightarrow 0} \frac{\sqrt{4+t} + \sqrt{4-t}}{2} \\
 &= \frac{\sqrt{4+0} + \sqrt{4-0}}{2} \\
 &= 2
 \end{aligned}$$

One sided limits

Ex

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \end{cases}$$

$$f(x) = \text{sgn}(x) = \frac{1 \times 1}{x} \quad D(f) = \mathbb{R} - \{0\}$$



Left limit  $\lim_{x \rightarrow a^-} f(x)$

x approaches a from the left



$$\lim_{x \rightarrow 0^-} f(x) = -1$$

$x$  approaches  $a$   
from the right

Right limit  $\lim_{x \rightarrow a^+} f(x)$

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

overall  
 $\lim_{x \rightarrow 0} f(x)$  D.N.E

theorem for  $\lim_{x \rightarrow a} f(x)$  to exist, the limits  $\lim_{x \rightarrow a^-} f(x)$ ,  $\lim_{x \rightarrow a^+} f(x)$  must exist and be equal

Example

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{|3x-1| - |3x+1|}{x}$$

$$|x| = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

if  $x$  is very small,  $3x-1 < 0$

$$= \lim_{x \rightarrow 0} \frac{|3x-1| - |3x+1|}{x}$$

$$= \lim_{x \rightarrow 0} \frac{(1-3x) - (3x+1)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{-\cancel{6x}}{\cancel{x}}$$

$$= -6$$

Ex Find  $\lim_{x \rightarrow 0^+} \sqrt{x^3 - x}$

① find domain of  $f(x) = \sqrt{x^3 - x}$

$\Rightarrow$  require that  $x^3 - x \geq 0$

$$x^3 - x = x(x^2 - 1) = x(x-1)(x+1) \geq 0$$

|            |           |      |           |     |           |     |           |
|------------|-----------|------|-----------|-----|-----------|-----|-----------|
| $x$        | $\ominus$ | $-1$ | $\ominus$ | $0$ | $\oplus$  | $1$ | $\ominus$ |
| $x+1$      | $\ominus$ | $0$  | $\oplus$  |     | $\oplus$  |     | $\oplus$  |
| $x-1$      | $\ominus$ |      | $\ominus$ |     | $\ominus$ | $0$ | $\oplus$  |
| $x(x^2-1)$ | $-$       |      | $+$       |     | $-$       |     | $+$       |

$$D(f) = [-1, 0] \cup [1, \infty]$$

Small positive values are not in the domain

$$\lim_{x \rightarrow 0^+} \sqrt{x^3 - x} \quad \text{D.N.E}$$

$$\lim_{x \rightarrow 0^-} \sqrt{x^3 - x} = 0$$

this one works  
small negative values are in the Domain

$$\lim_{x \rightarrow 0^+} \sqrt[3]{x^3 - x} = 0$$

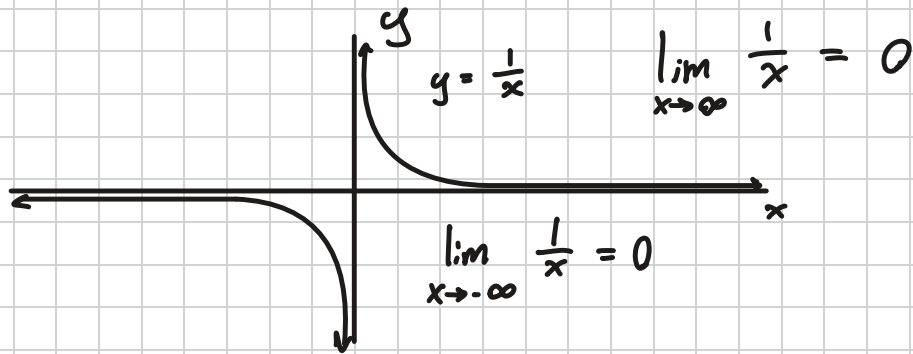
$$\lim_{x \rightarrow 0^-} \sqrt[3]{x^3 - x} = 0$$

} No domain problems

# Limits at infinity

$\lim_{x \rightarrow a} f(x)$  a finite

Consider  $f(x) = \frac{1}{x}$   $D(f) = \mathbb{R} - \{0\}$

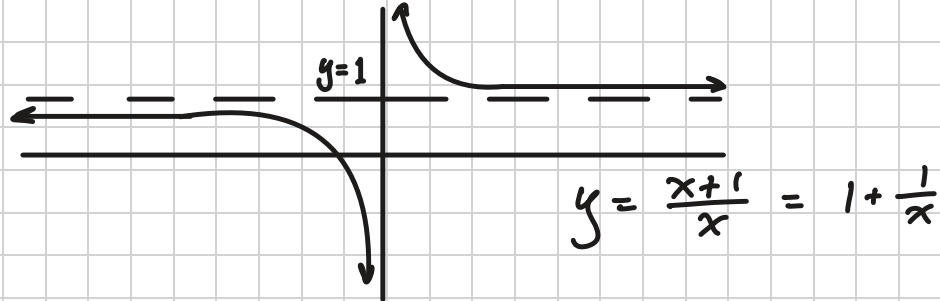


Ex  $f(x) = \frac{x+1}{x}$

$$\lim_{x \rightarrow \infty} \frac{x+1}{x}$$
$$= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)$$

$$= 1$$

Same thing for  $\lim_{x \rightarrow -\infty} \frac{x+1}{x} = 1$



Horiz Asymptote @  $y=1$

for  $y = \frac{1}{x}$ , Asymptote @  $y=0$

Ex  $\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$

in general

$$\lim_{x \rightarrow \infty} \frac{1}{x^a} = 0, \quad a > 0$$

Ex

$$\lim_{x \rightarrow \infty} \frac{5x^3 + 3x - 7}{8x^3 + x^2 - 1}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{5x^3 + 3x - 7}{x^3}}{\frac{8x^3 + x^2 - 1}{x^3}}$$

$$= \lim_{x \rightarrow \infty} \frac{5 + \frac{3}{x^2} - \frac{7}{x^3}}{8 + \frac{1}{x} - \frac{1}{x^3}} = \frac{5}{8}$$

*(Note: Red circles and arrows in the original image highlight terms that approach 0 as x goes to infinity: 3/x^2, 7/x^3, 1/x, and 1/x^3.)*

$$\begin{aligned}
 \underline{\underline{Ex}} \quad & \lim_{x \rightarrow \infty} \frac{2x+7}{x^2-11} \\
 &= \lim_{x \rightarrow \infty} \frac{\left(\frac{2}{x}\right)^{\rightarrow 0} + \left(\frac{7}{x^2}\right)^{\rightarrow 0}}{1 - \left(\frac{11}{x^2}\right)^{\rightarrow 0}} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \underline{\underline{Ex}} \quad & \lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{3x^2-x+2}} \\
 &= \lim_{x \rightarrow -\infty} \frac{\frac{2x-1}{x}}{\frac{\sqrt{3x^2-x+2}}{x}} \\
 &= \lim_{x \rightarrow -\infty} \frac{2 - \frac{1}{x}}{-\sqrt{\frac{3x^2-x+2}{x^2}}} \\
 &= \lim_{x \rightarrow -\infty} \frac{2 - \frac{1}{x}}{-\sqrt{3 - \frac{1}{x} + \frac{2}{x^2}}} \\
 &= -\frac{2}{\sqrt{3}}
 \end{aligned}$$

Note :  $\sqrt{x^2} = |x|$   
 if  $x < 0$   
 $\sqrt{x^2} = -x$

$$\underline{\underline{Ex}} \quad \lim_{x \rightarrow \infty} \frac{2x-1}{\sqrt{3x^2-x+2}} = \frac{2}{\sqrt{3}}$$

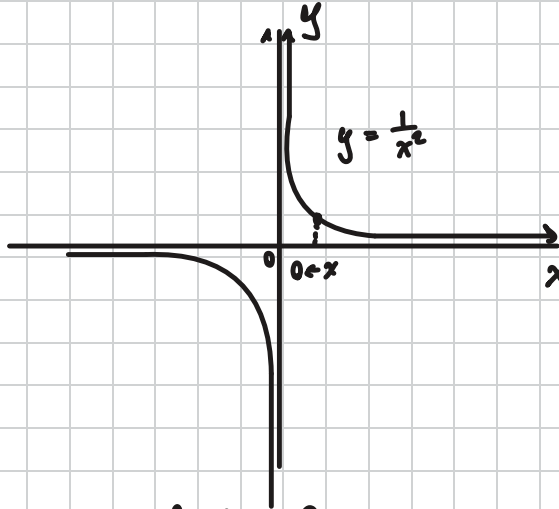
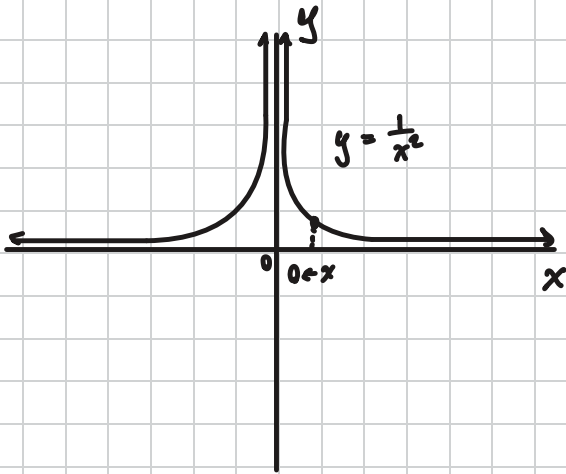
Ex

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) \\
 &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x} - x)(\sqrt{x^2 + x} + x)}{\sqrt{x^2 + x} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{\cancel{x^2} + x - \cancel{x^2}}{\sqrt{x^2 + x} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{x}{x}}{\frac{\sqrt{x^2 + x}}{x} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{\frac{x^2 + x}{x^2}} + 1} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{2}
 \end{aligned}$$

# infinite limits

$$f(x) = \frac{1}{x^2}$$

$$D(f) = \mathbb{R} - \{0\}$$



$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ DNE}$$

(a finite)

$$\text{if } \lim_{x \rightarrow a^+} f(x) = \pm \infty$$

$f$  has a asymptote

Example

$$\lim_{x \rightarrow 0^+} \frac{x+1}{x}$$

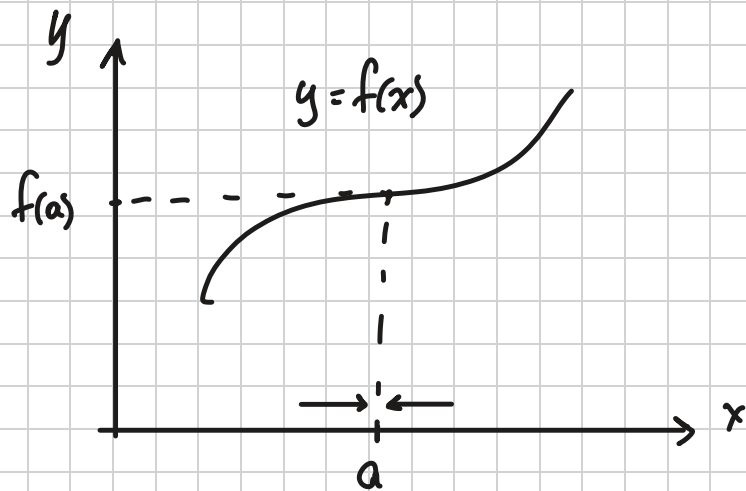
$$= \lim_{x \rightarrow 0^+} \left( 1 + \frac{1}{x} \right) = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{x+1}{x}$$

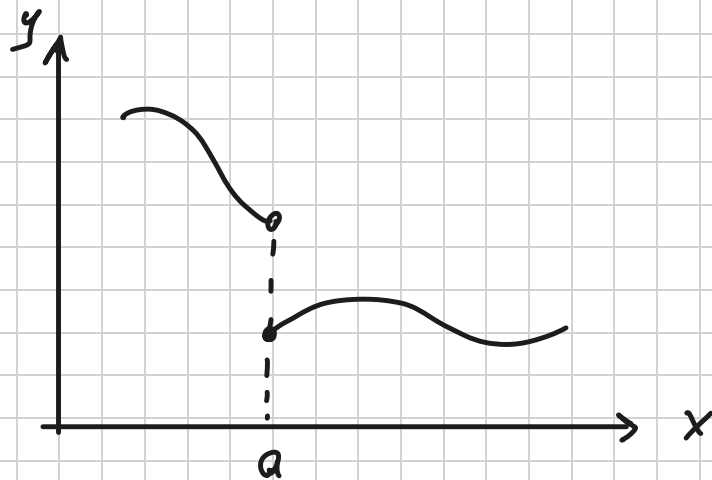
$$= \lim_{x \rightarrow 0^-} \left( 1 + \frac{1}{x} \right) = -\infty$$

$$\lim_{x \rightarrow \infty} x = \infty, \quad \lim_{x \rightarrow \infty} x^a = \infty, \quad a > 0$$

# Continuous functions



Continuous



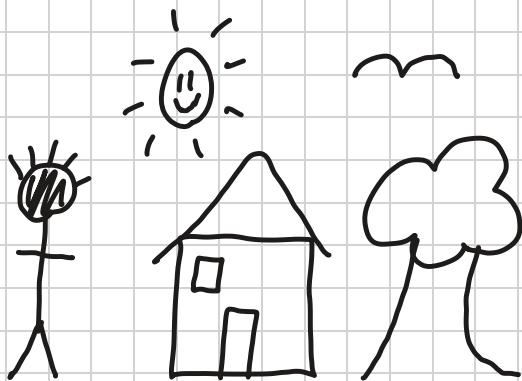
discontinuous

Def  $f$  is Continuous at  $a \in D(f)$

$$\text{if } \lim_{x \rightarrow a} f(x) = f(a)$$

$f$  is Continuous on an interval  $I \subset D(f)$

if  $f$  is Continuous at every point  $a \in I$



Def  $\Rightarrow f$  is right continuous at  $a \in D(f)$

$$\text{if } \lim_{x \rightarrow a^+} f(x) = f(a)$$

$\Rightarrow f$  is left continuous at  $a \in D(f)$

$$\text{if } \lim_{x \rightarrow a^-} f(x) = f(a)$$

for a function to be continuous at a  
it must

Ex 18  $g(x) = \begin{cases} x-m, & x < 3 \\ 1-mx, & x \geq 3 \end{cases}$

Find values of  $m$  such that  $g$  is continuous everywhere.

$g$  must be continuous at  $x=3$

$$\lim_{x \rightarrow 3^+} g(x) = 1-3m, \quad \lim_{x \rightarrow 3^-} g(x) = 3-m$$

$$g(3) = 1-3m$$

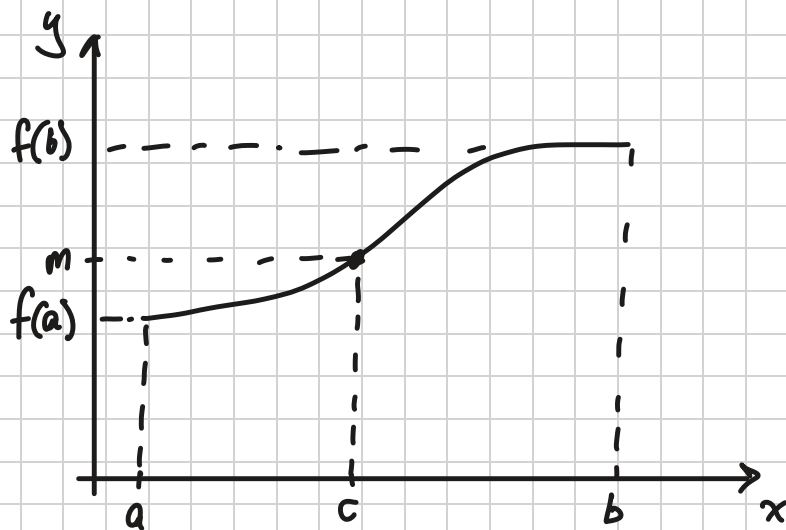
we must have  $1-3m = 3-m$

$$3m-m = 1-3$$

$$2m = -2$$

$$m = -1$$

# intermediate value theorem



$f$  continuous  
on  $[a, b]$

if  $m$  is such that  $f(a) \leq m \leq f(b)$

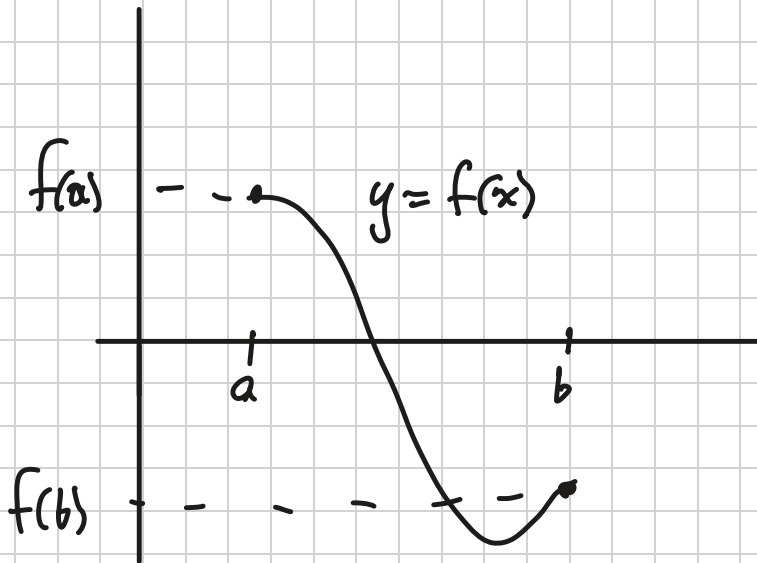
$\Rightarrow$  there is a point  $c$ ,  $a \leq c \leq b$

such that  $f(c) = m$

Ex  $f(x) = x^3 - 15x + 1$

$$D(f) = \mathbb{R}$$

prove that the equation  $f(x) = 0$   
has 3 solutions in  $[-4, 4]$



$f(a)$  and  $f(b)$  opposite signs

$\Rightarrow$  there is  $c$ ,  $a \leq c \leq b$

such that  $f(c) = 0$